Code No: 117EF **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
MACHINE LEARNING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Define Concept Learning.	[2]
b)	What are the advantages of PAC model?	[3]
c)	Define neural network learning.	[2]
d)	Write about learning with active queries.	[3]
e)	What is over fitting?	[2]
f)	What are the advantages of hidden markov models?	[3]
g)	Define case based reasoning.	[2]
h)	Write about lazy based rule learning.	[3]
i)	Define explanation based learning.	[2]
j)	What is the role of genetic program in machine learning?	[3]

PART – B

(50 Marks)

2.a)	Discuss Version Space Method in Learning Algorithm.	
b)	What are the issues in machine learning?	[5+5]
	OR	
3.a)	Discuss the approaches followed to complete a learning tasks.	
b)	What are the issues regarding data source and success criteria?	[5+5]
4.a)	Explain the concept of pruning of decision tree.	
b)	How we can support the Occam's razor as an inductive bias in machine learning.	[5+5]
	OR	
5.a)	Why is Backpropagation Required? What are the advantages and limitations of Backpropagation?	
b)	How we can support the Occam's razor as an inductive bias in machine learning.	[5+5]
6.a)	Explain the various methods to perform cross validation.	
b)	What are the multiplicative rules for weight tuning?	[5+5]
	OR	
7.a)	Define sampling theory and write the basics of sampling theory.	
b)	Write about brute force bayes concept learning.	[5+5]

8.a) Explain case based learning technique.
b) Compare lazy and eager generalization methods. [5+5]

OR

9.a) Write the K nearest neighbor algorithm.
b) Discuss in brief instance based techniques. [5+5]

10.a) Discuss different search methods for induction.
b) Write about advantages and disadvantages of Genetic algorithms. [5+5]

OR

11.a) Illustrate genetic operators.
b) How can sample complexity be reduced in genetic programming? [5+5]

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